

Words by [Ben Hopkinson](#)

How Madrid built its metro cheaply

5th December 2024



23 Mins 

Madrid tripled the length of its metro system in just 12 years — faster and cheaper than almost any other city in the world. What can its expansion teach other cities?

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growing spree. Over the course of the next 12 years, the metro grew by 126 miles (203 kilometers), nearly tripling in length. This expansion made it one of the world's fastest-growing metros, on par with Beijing and Shanghai. Today, Madrid has the sixth longest metro outside of China, and the third longest in Europe after London and Moscow.

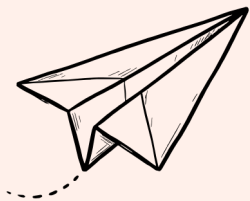
Madrid's metro system (1994)



Source: Wikimedia

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Madrid's metro system (after 2007)



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The 35-mile (56 kilometer) program of expansion between 1995 and 1999 cost around \$2.8 billion (in 2024 prices). New York's 1.5-mile extension of the 7 subway to Hudson Yard cost about the same (adjusted for inflation). London's Jubilee Line Extension, built at the same time as Madrid's expansion, cost nearly ten times more per mile than Madrid's program. The World Bank described Madrid's costs as 'substantially below the levels that were internationally considered possible'. Since the 1990s, Madrid, and Spain as a whole, has continued to build infrastructure at some of the lowest costs in Europe.

Subway costs in selected cities since 1995

Project title	City	Cost (inflation-adjusted)	▲ Cost per mile	Length
1995–1999 Programme	Madrid	£2,243M	£64M	35 n
1999–2003 Programme	Madrid	£3,348M	£72M	47 n
2003–2007 Programme	Madrid	£4,096M	£91M	45 n
Silver Line	Washington	£5,455M	£237M	23 n
Green Line Extension	Boston	£2,069M	£438M	5 n
University Link	Seattle	£1,468M	£463M	3 n
Broadway Extension–Millennium Line	Vancouver	£1,966M	£555M	4 n
Jubilee Line Extension	London	£6,049M	£609M	10 n

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Longo to Richmond Hill	Toronto	£3,778M	£822M	5 n
Ontario Line	Toronto	£12,884M	£1338M	10 n
Crossrail	London	£18,157M	£1392M	13 n
BART to Silicon Valley	San Jose	£9,893M	£1649M	6 n
7 extension	New York	£2,196M	£2209M	1 n
Second Avenue Subway	New York	£9,555M	£2903M	3 n

Note: costs are given in 2023 GBP.
Source: [Hopkinson & Dumitriu \(2023\)](#)

Madrid’s success provides four key lessons for policymakers and engineers in places that struggle to cheaply build new transit.

- City-level powers rewarded fast, inexpensive delivery. The structure of the Community of Madrid concentrated the

planning, funding, and construction powers at the right level to deliver the project. This enabled political entrepreneurs to make electoral promises about delivering new infrastructure and have their political fortunes dependent on success.

- Time is money. The regional government streamlined environmental and planning processes and the company that oversaw construction expedited the building by tunneling 24/7.
- Trade-offs matter and need to be explicitly considered. The metro planners recognized the trade-offs that exist between station design and cost, signaling complexity and how much testing is required, and tried-and-tested technology versus innovation.
- A pipeline of projects enables investment in state capacity. Madrid built the necessary state capacity to deliver the project, with experienced engineers and managers working in-house to deliver the technical design and oversee construction. The public company tasked with construction could pay extra to hire experts and procure based on cost and quality instead of just

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Following the restoration of democracy in Spain after decades under Francisco Franco, the 1978 constitution created 17 autonomous communities. These function like an American or Canadian federal state, and include the Basque Country, Catalonia, Andalusia, Madrid, and so on. Each of them has an elected parliament, which in turn chooses the President of the Community (a bit like the UK Parliament in Westminster). The Community of Madrid includes the city of Madrid and other outlying cities and towns and is a little larger in size than the US state of Delaware and a little smaller than Île-de-France, the French region including Paris.

Some countries, like Britain, have a centralized system, where the approval of new local transport projects is at the discretion of national ministers and Parliament, and funding for construction mostly comes from the Treasury and national taxes. This system creates friction as local authorities have to continually ask the

central government for permission to build and for funding. Once they’ve been given funding, there are limited incentives to keep costs down as local leaders are spending someone else’s money on construction.

The Community of Madrid demonstrates a more successful structure. Madrid’s regional assembly has high flexibility in levying taxes, including income and VAT, approves a roughly €25 billion budget (68 percent higher than London’s budget per capita), and is in control of the Regional Consortium of Transportation for Madrid, much as the state of New York controls the Metropolitan Transit Authority (MTA). The regional consortium, in turn, funds and oversees the Madrid Metro, light rail, and urban buses.

The winner of regional assembly elections has all of the levers of control over a project at their disposal. They can approve new projects, fund those projects by borrowing, and oversee the construction to deliver the project. Enterprising politicians at the regional level of government could claim that they are going to

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smaller than the area the project affects, as in Britain and America, it is harder for one politician to take ownership over the construction of a project. This limits the accountability of those in charge if something goes wrong, weakening the incentives and ability of politicians to take full responsibility for successfully delivering the project.

The beginning of the metro’s rapid growth happened in the run-up to the 1995 regional assembly election. The election pitted the incumbent left-leaning Spanish Socialist Workers’ Party (PSOE) against the center-right People’s Party (PP). Instead of debating whether or not to expand the metro, the two parties competed on how much they could feasibly build within the next four-year term.

The PP candidate for Assembly President, Alberto Ruiz-Gallardón, promised to deliver 30 new miles (48 kilometers) of metro by the next election, compared to the 14 miles (23 kilometers) the PSOE had delivered during their previous 15 years in government. With

this pledge, the PP won a majority and Ruiz-Gallardón was duly elected president.

Madrid’s four-year electoral cycles gave tight timelines for projects being done. Yet with strong political commitment and the knowledge that fulfilling headline electoral promises would be rewarded by voters, extensions were delivered within four-year windows.

Throughout construction, community engagement remained limited and top-down. The metro planners held public meetings, but they were informational rather than consultative. Some feedback was solicited, but this generally involved the location of entrances of stations, not whether the line should go ahead or where it should run to. If a majority of the residents affected by some sub-element of the program, such as the location of an entrance to a station, wanted a change to the plans, they could be modified. But if the change entailed an increase in costs, project staff were likely to deny the request

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next expansion should go further: close to their workplace or their home, everyone wanted to feel involved’. The principle behind this approach to consultation was that the government (and its recent election promises) represented the democratic will of the community, as opposed to that of a vocal minority who might show up at several rounds of consultation to oppose plans.

The timelines involved in the metro construction were politically fortuitous to the PP. Each metro expansion created a cycle that enabled the PP to open the lines and stations right before the next election. These events were demonstrations of the PP’s success, which made them useful political talking points in the run-up to the regional elections, where the PP would promise further expansions of the metro to currently underserved areas of the community. Ruiz-Gallardón described the feeling of Madrid residents after the first four years of building as: ‘it was the citizens themselves who recognized that it had been worth it and asked us to continue on [building].’

Voters responded positively to the PP’s plans to build new metro lines to their areas. For example, in the run-up to the 1999 elections, the PP pledged to construct a circular metro line between five historically working-class cities to Madrid’s south. While it improved their percentage of the vote by just 0.1 percentage points across the Madrid community, they increased their share of the vote by 3 percentage points across those five cities that would be connected by MetroSur, despite them being traditional working-class PSOE strongholds.

On its own, having democratic accountability and the levers of planning, funding, and construction at the right level is not a guarantee of low costs. There are some American projects done with planning and funding at the right level, such as the D Line Extension in Los Angeles, that are more expensive than global averages. But the American projects that are self-initiated, self-directed, self-funded, self-approved, and in politically competitive jurisdictions do better. For example, Portland, Oregon’s streetcar

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LESSON 2: TIME IS MONEY

There’s an old idea in project management of the restrictions facing projects that can be summed up as, ‘good, fast, and cheap: choose two’. If you want to have a quality final product, the management team has to choose between doing it cheaply but taking a long time and choosing to do it fast but with a lot of extra expense.

This line of thinking is present in many British and American infrastructure projects. For example, when the cost of California’s high-speed rail project kept rising, project managers decided to cut parts of the program that ran into San Francisco and Los Angeles and instead focus on building the core section between Merced and Bakersfield, with future phases actually connecting the main cities. In this case, the project team chose to slow down delivery to save on cost.

However, Madrid’s example shows that this trilemma isn’t an iron law of nature: it is possible to build a high-quality end result both cheaply and quickly. By building quickly there is less time when the project is exposed to global shocks, inflationary pressures, or chopping and changing by different politicians, which all drive costs up. To speed up building, Madrid removed burdensome environmental impact requirements, sped up the approval of projects, and worked round the clock to finish construction.

The Madrid Metro handled environmental impact assessments briskly. The assembly’s environmental evaluation law allowed impact assessments to be abbreviated if the extension was in an already established urban area, as the impacts to nature would be lower. This abbreviation meant that the project had to complete fewer requirements, such as studies on increased carbon emissions or the impact on farmland. The metro projects completed a streamlined assessment with the required parts clearly laid out in law, and would get confirmation within five months, compared to

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removal, and environmental surveillance that were easily met. Contrast this with the 3.3-mile (5.3-kilometer) Portishead branch line reopening in the South West of England, which had a 17,912-page-long environmental statement. On a per-mile-of-new-track basis, Portishead’s was 1,142 times longer than Madrid’s. While the costs of Madrid’s environmental monitoring plan were assessed and controlled, in Britain environmental mitigations can run into the hundreds of millions. As part of building the high-speed railway HS2, the British built a kilometer-long tunnel to protect bats, at a cost of more than £100 million, despite there being no evidence that bats were at risk from the trains.

The assembly accelerated permitting and removed multiple levels of review. While in Britain, new local transport projects have to go through a three-or-more-year review by the national Department for Transport and get the Transport Secretary’s sign off, the Madrid Metro projects faced much lower hurdles. The regional transit authority and the Madrid Metro company identified areas that

could be better served by a metro expansion and then created expansion plans. These plans were then submitted to the Government of the Community, where they were evaluated and approved by the Community’s Department of Public Works, Urbanism, and Transport, which was led by Luis Eduardo Cortés, the Vice President of the Community.

The process was unified, with the regional government and transit authority working together to secure approvals. Since the PP had made metro expansion one of their biggest electoral promises, they were conscious that they would be judged on whether they had delivered at the next election. Decisions requiring political input, such as station locations, were often made within 24 hours, compared to months or even years in other countries. For example, design changes and safety improvements to Green Park station as part of the London’s Jubilee Line Extension had to go through Parliament as a bill that was introduced in 1991 , but didn’t become an act until 1994.

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was only able to tunnel for 13 hours a day in some areas to allay residents’ noise concerns. Manuel Melis Maynar, the head of the Madrid Metro during the expansion program explained the straightforward discussions he had with residents to get approval over the 24-hour working schedules: ‘In Madrid, everyone understood . . . that there would be disruption on the streets for three or four months, but it was for their benefit in that they would have a beautiful metro system. Everyone was supportive and understood that we had to divert traffic and close streets.’

By explicitly tying the benefits of going quickly with its reduced construction time and disruption, the Metro leadership was able to convince the public to accept the 24-hour working schedules. The fast pace of delivery also meant that Madrid residents saw the benefits of new metro lines and stations opening up quickly, which increased the support of future construction works. Seeing how their daily lives could improve, residents would advocate for new lines near them.

Lesson 3: Trade-offs matter and need to be explicitly considered

Building a new metro line, or any other large infrastructure project, requires politicians and project managers to grapple with trade-offs between design features and cost. In Britain and America, new infrastructure projects often come with promises that they will be the fastest, grandest, most innovative designs.

New York’s recent Second Avenue subway suffers from grandiose designs taking priority over delivering simple, replicable stations. The NYC Department of Transport mandated at least four lanes of traffic must stay open throughout construction, limiting cheap cut-and-cover station construction. New York City Transit requested extensive storage, utility, and personnel rooms in the underground stations. They could ask for these rooms because they weren’t the ones paying for them and the MTA had to give in because NYC Transit could withhold approvals if they didn’t get what they

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Former Madrid Metro boss Manuel Melis Maynar’s advice that ‘design should be focused on the needs of the users, rather than on architectural beauty or exotic materials’, wasn’t followed by London’s Jubilee Line Extension, built at the same time as the Madrid Metro extensions. Each of its stations had a different architect, creating large, complex designs with vast chasms between the surface and the tracks. While these were popular with architecture critics – several stations were shortlisted for Britain’s most prestigious architecture award, the RIBA Stirling Prize – they were also expensive to construct. London Bridge’s Jubilee line station cost £217 million in 2024 terms.

Madrid took a different approach. As far as possible, stations were standardized, using copy-and-paste designs. This simplified the engineering and design work required, while construction crews would gain experience and efficiency in building stations. The new stations are simple and replicable and are also easy to navigate.

Busier stations have sculptures and murals to add character, a far cheaper way to spruce up a station than building an architectural behemoth. While per-station costs aren't available, even if the entire budget of the 1995–1999 program was only spent on stations, they would still be just £60 million each in 2024 terms, less than a third as much as London Bridge's Jubilee line station.

The local funding of the project, and the electoral consequences of budget overruns and failure to deliver both make it more likely that a project will choose to pick the cheaper options, rather than the expensive nice-to-haves.

Trade-offs exist when it comes to signaling too. The planners could have opted for moving block signaling, where computers calculate the exact location of trains in real time to determine safe distances between them. While this brings benefits in capacity, it is also more expensive and much more complicated to build and test. Instead, the planners chose to use tried-and-tested fixed block signaling, which divides the track into blocks where there could be

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and the two Victorian railways to the east and west of the city center. This complexity required three years of testing with trial trains running through the tunnels. In Madrid, by contrast, the window between the end of construction and the opening of the line was just three weeks. While signaling innovation will serve the Elizabeth line well in the long run, the delays increased the cost by two billion pounds, about the same as the cost of the entire 1995–1999 Madrid program.



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Allegory of a Tunnel Boring Machine by Coco Ortega in
Móstoles Central

IMAGE
Source: Author's collection

Lesson 4: A pipeline of projects enables investment in state capacity

The Community Assembly, led by Alberto Ruiz-Gallardón, set the strategic direction for the project and hired a team of nine engineers to oversee it. Unlike infrastructure projects in Britain or America, which are heavily reliant on external consultants to handle all stages of the project, this group of well-paid in-house engineers led much of the Madrid Metro expansion. The team stayed largely the same throughout the different projects, meaning that they were able to learn from their experience and apply it to future projects.

One of those experts, the former head of the Metro, Manuel Melis Maynar, wrote that, ‘a well-executed tunneling project is an art. The client should be prepared to spend the necessary time in choosing the artist.’ The senior project team included four geotechnical experts who conducted in-depth soil investigation and monitoring to determine which tunneling method would be most suitable for each section.

Unexpected challenges in tunneling, such as soil compositional changes or a collapse, are one of the biggest risks involved with metro construction. Spanish engineers had their eyes on London, where in 1994, construction of a new rail link to Heathrow Airport went disastrously wrong when a new form of tunneling led to a collapse of two tunnels. The UK Health and Safety Executive described it as ‘the worst civil engineering disaster in the UK in the last quarter century’. This led to costs spiraling to £150 million (£305 million today), three times over budget, and added two years of extra delays.

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experience. According to project folklore, these engineers would meet up after their shifts in Madrid’s tapas bars and discuss their challenges or successes, engaging in informal competitions to tunnel faster, encouraging speed and reducing costs. With the pipeline of projects, the metro leadership could then rehire the teams that were working best for future projects.

All of the expertise and lessons was captured by Mintra, which was a public company that the regional assembly established to oversee the 1999–2003 and 2003–2007 programs of work. Mintra drew up and managed contracts. The contracts had fixed lump sum payments, but also a bill of quantities (i.e., a list of agreed prices for any extra jobs contractors could be asked to do), so that any additional work could be readily priced and agreed to with contractors.

This avoided the potential for litigation, mediation, and work stoppages that have often occurred on British infrastructure

projects, which try to agree on a fixed price for the entire job to offload risk to the private sector. But it also avoids the problems with cost-plus contracts (like regulated asset base), where contractors have no incentive to constrain spending because they are paid an agreed markup on top of their costs. When unexpected challenges occur, as was the case with Edinburgh’s tram project, disputes over payment often follow. These can quickly shut down worksites, causing the whole project to grind to a halt.

Mintra’s procurement strategy focused on value for money rather than just cost. The bid price from the company was weighted to be only 30 percent of the evaluation, unlike in the US and UK where it is often the primary or only consideration. Edinburgh’s Tram and Boston’s Big Dig were both overly focused on the lowest-bid contractors in a misguided attempt to offload risk to the private sector, which ultimately led to change orders, delays, and quality issues. In Madrid, the remaining 70 percent was split between 20 percent for the project time and 50 percent for the technical merits

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Mintra had an added accounting benefit too. It was a private entity so its debt was not on the government’s books, but it was owned by the regional government, so investors had the confidence of government backing, enabling Mintra to borrow at competitive rates. Mintra was also not tied to regional government pay scales, so it could pay more to bring in and retain highly skilled engineers and planners. Once the metro expansions were done, Mintra would transfer the completed tracks over to the Madrid Metro company, while the Community of Madrid continued to pay off Mintra’s debt through general taxation.

The regional government and Mintra were able to build up capacity to oversee the metro expansions and capture lessons on how to efficiently build new lines to be used for future projects. The tunneling, contracting, and project management expertise drove the project to meet tight deadlines and keep costs lower than anyone had thought possible.



Jorge Bernabeu's Landscapes of the South in Puerta del Sur.

IMAGE

Source: Author's collection

Madrid today

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Government decided to a plan on government bonds, which caused expensive debt and painful austerity. Unsurprisingly, this meant there was no money from either the government or private capital for further expansions. But by this point, the Madrid Metro was already among the most comprehensive underground systems in the world.

Today the Madrid Metro, and all of its expansions, prove to be very popular amongst madrileños. In the most recent ridership survey, 75 percent of passengers described themselves as 'very satisfied', giving the metro an average rating of 8.2 out of 10. That rises to 8.9 out of 10 for foreign visitors to the city. By comparison, New York's subway had a paltry 56 percent satisfaction rate in the most recent survey.

The level of satisfaction is seen in ridership numbers too. The Madrid Metro carried 677.5 million riders in 2019, equating to 210 trips per resident. That beats both London Underground's 155 trips per Londoner and the NYC subway's 203 trips per New Yorker.

MetroSur, the line that links five smaller cities to Madrid’s south, has seen almost a billion trips made since it opened. Per year, just this one line would be the sixth busiest American subway system, carrying more passengers than the metros of San Francisco, Philadelphia, Atlanta, or Miami.

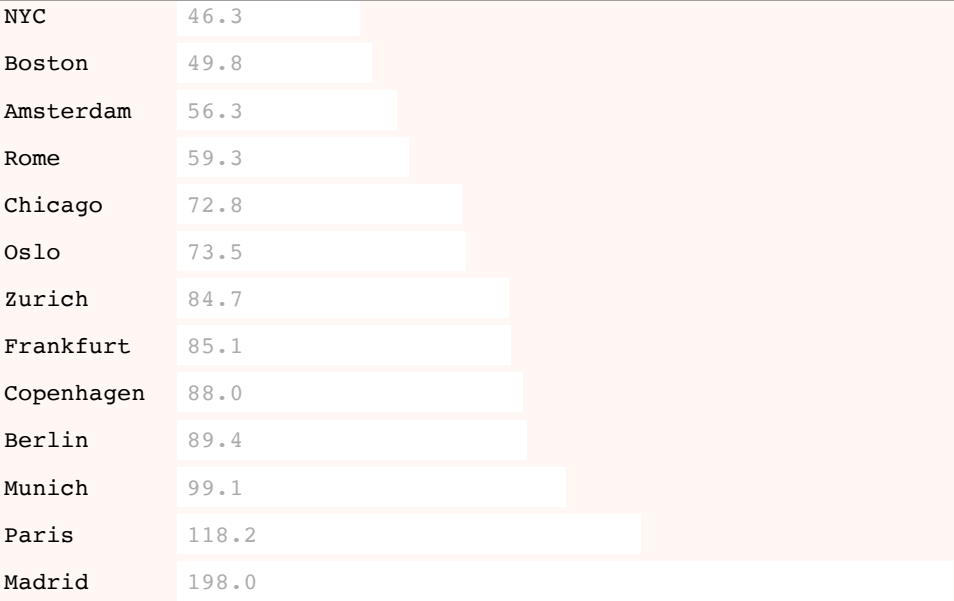
The expansion had clear economic benefits. One study found that houses that were nearby stations on the MetroSur expansion were worth eight percent more than houses not within walking distance. Another found that firms, especially retail businesses, were likely to cluster around MetroSur stations. Out of the main American and Western European cities, Madrid has the largest area accessible within 30 minutes and 45 minutes by public transport, which increases the size of the labor market as more people can easily get to and from the most productive areas of the city.

Area accessible by public transport within 30 minutes

Square kilometers accessible within 30 minutes of the

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Madrid’s metro expansion provides plenty of lessons for cities that aspire to improve their transport network. If the anglosphere could build at Madrid’s cost per mile, cities could be transformed. Instead of stopping at Crossrail 1, London would be charging into Crossrails 2, 3, and 4. New York would be able to build the Interborough Express between Brooklyn and Queens and run

subway trains to LaGuardia Airport. Dublin, Adelaide, and Ottawa could build metros. Leeds, Wellington, and Christchurch could all get a light rail system.

These advances won’t happen overnight and there will always be challenges when building metro systems. But perhaps the most important lesson of all in Madrid’s metro expansion is for our civil servants, engineers, and politicians to consider how other places deliver large infrastructure projects. By sharing best practices and engaging with successful cases, we can learn and improve the way we build.



Ben Hopkinson is head of research at Britain Remade, where he works on policy development and analysis. You can follow him on Twitter.

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